

An aerial photograph of a city grid is shown. Overlaid on the right side of the image is a 3D bar chart. The bars are colored in a gradient from purple to yellow, with the tallest bars being yellow and orange. The bars vary in height and are arranged in a way that suggests a data visualization of urban density or building heights.

BPS5231

Artificial Intelligence

For

Sustainable

Building

Design

Ang Yu Qian

Assistant Professor



How can we leap past today's limits to develop climate-resilient, human-centric buildings and cities? This studio-lecture hybrid equips participants with a suite of AI workflows – data-driven simulation, generative AI, deep-learning surrogates, and large-language-model copilots – applied through a series of provocative weekly “Design Beyond” lenses. Design-and-code labs, expert critiques, and a capstone project that demonstrates how AI can move sustainable design well beyond today's boundaries.

Welcome Message



L01.1

Introduction

First things first

Course Details

Who are we?

Who are you?

Class Organization

L01.2

Basics

Buildings

History of AI

Machine Learning Basics

L01.3

Applications

Examples

Design Beyond Space

L01.1

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L01.3

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Design Beyond Space

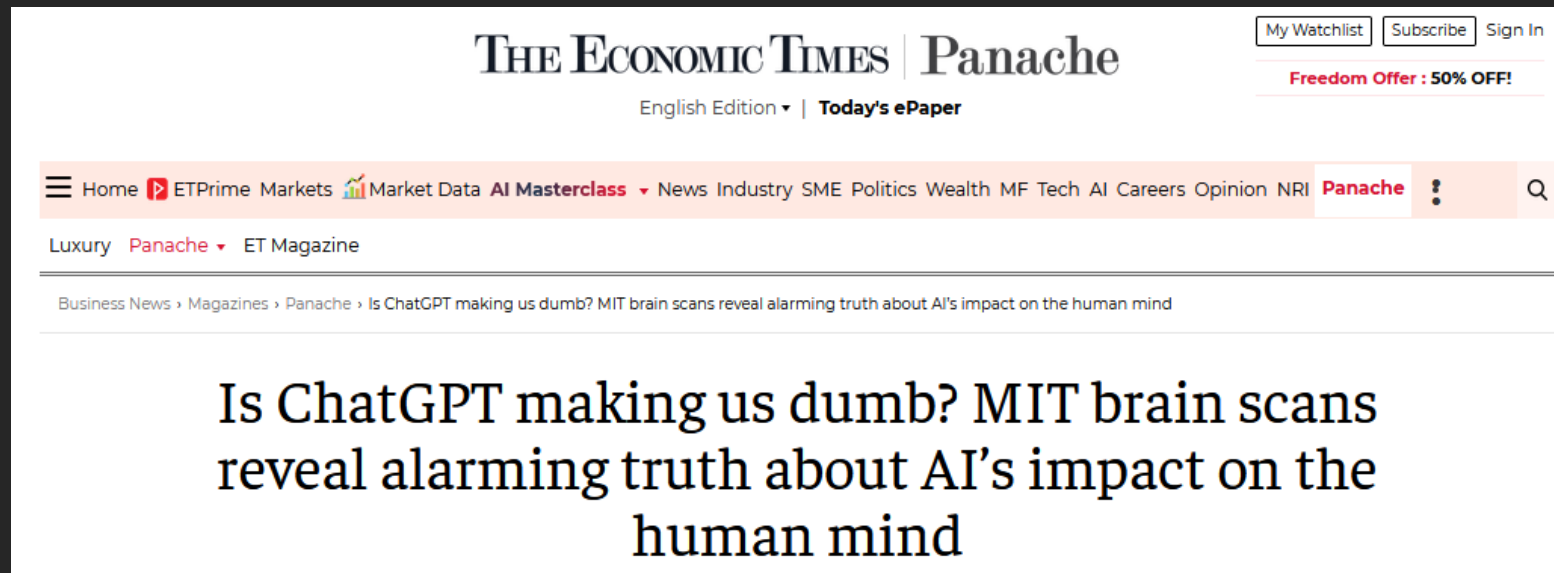
First things first

I am not admin

I am also not IT

You will need to code

Vibe coding is fine, this is an AI course, but you need to know what the model is generating. Repeatedly using ChatGPT/Claude/DeepSeek without thinking will make you stupid



First things first

We will be doing
modeling & simulations

First things first

This will be hacky

It's a new course. It's AI. Things will go wrong.

Who are you?

Go to
<https://bit.ly/BPS5231-class>



Who are we?

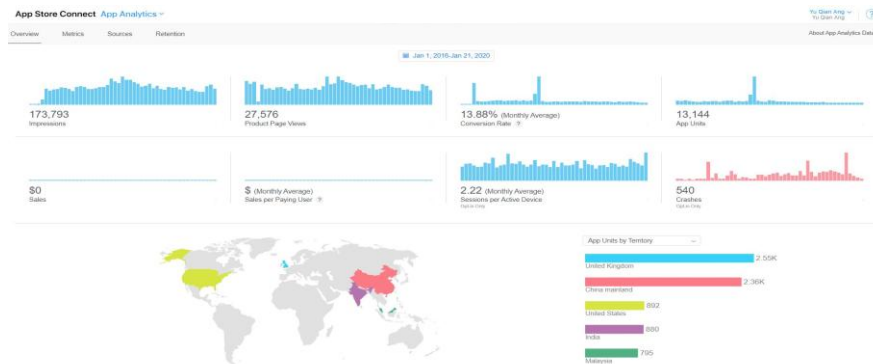
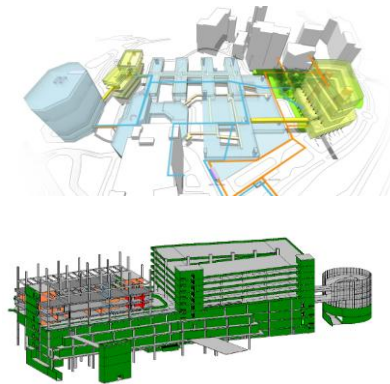
National University of Singapore

B.Sc. Project & Facilities Management



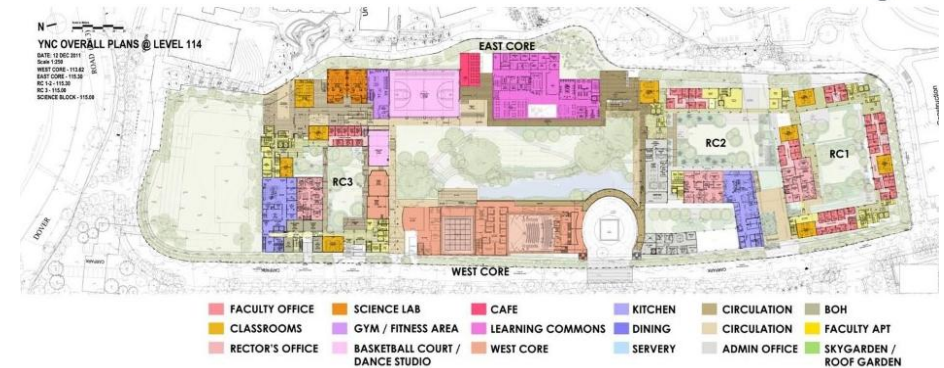
National University of Singapore

M.Sc. Building



Rider Levett Bucknall

Consultant + BIM Lead



INTRODUCTION

BASICS

APPLICATIONS

Singapore Ministry of Trade & Industry Senior Assistant Director



University of Cambridge Gates Scholar, MPhil Technology Policy



Massachusetts Institute of Technology

PhD



Harvard University
Visiting Fellow



The World Bank Group
Consultant



NUS City Syntax Lab
Assistant Professor



Urbanflow
Cofounder



City Syntax Lab



Shuyang Zhang
Research Fellow



Ruohan Xu
Research Fellow



Wanyu Pei
Research Fellow



Lujia Zhu
PhD Student



Sparsh
Undergraduate Researcher &
Founder (Thermal Floater)



Jiajing Chew
Undergraduate Researcher



Wang Yunjia
Graduate Research
Assistant



Ryan Tan
Cofounder & Software
Engineer (UrbanFlow)



Bryan Ong
Cofounder (UrbanFlow)



Peining Wong
Design Lead (UrbanFlow)



Hueyu Chen
PhD Student



Ye Xia
PhD Student (Innovation
Track)



Liren Wang
PhD Student



Yazhe Liu
PhD Student



Faizatu Zahrah
Rahmaniah
PhD Student (Industry
Track - Co-Supervised with
Dor & WWH Architects)



Coco Li
Undergraduate Researcher
(UrbanFlow)



Shana Soh
Student Software Engineer
(with BCR)



Justine C. Lim
Graduate Research
Assistant



Wenqing Dong
Graduate Research
Assistant



Yudian Chen
Graduate Research
Assistant



Lester Ong
Graduate Researcher/
Developer



Royden Soh
Undergraduate Researcher



Azaan Oureshi
Undergraduate Researcher



Jolyn Teo
Undergraduate Researcher



Eddison Chung
Undergraduate Researcher



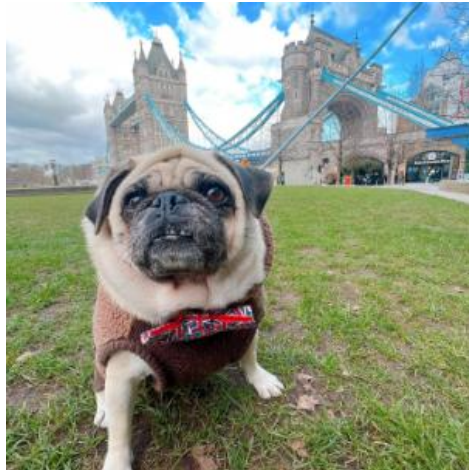
Minrui Zhang
Graduate Research
Assistant



Lab Pug
also Dogtural Candidate

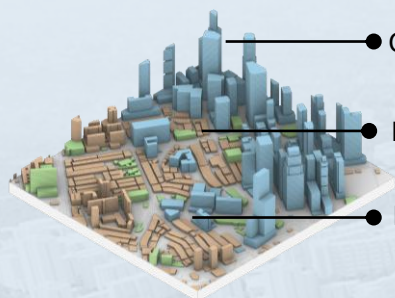


Yu Qian Ang
PI / Presidential Young
Professor



@breadloafmonster

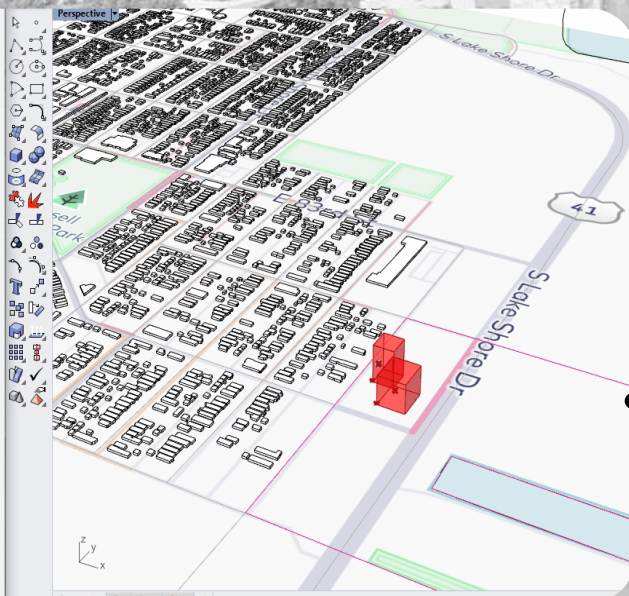
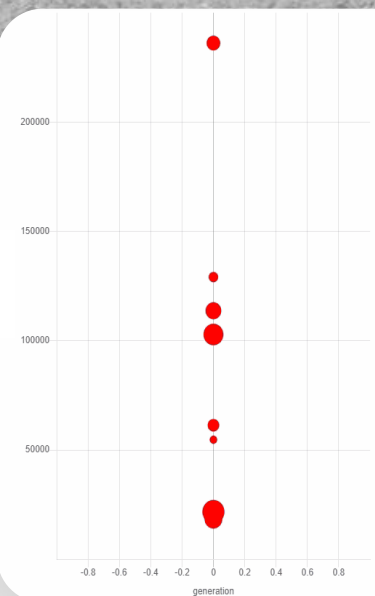
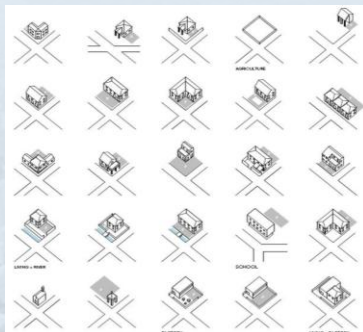
Simulating a thousand cities



Commercial

Public

Residential

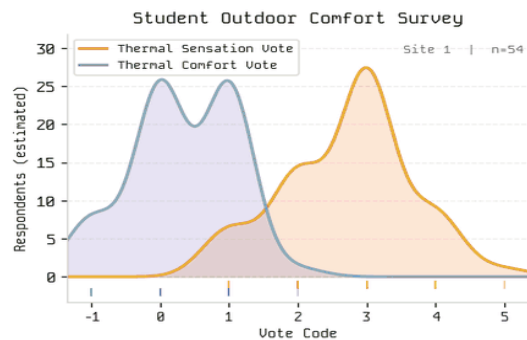
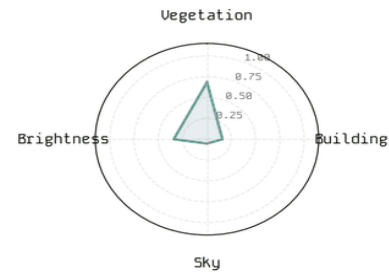
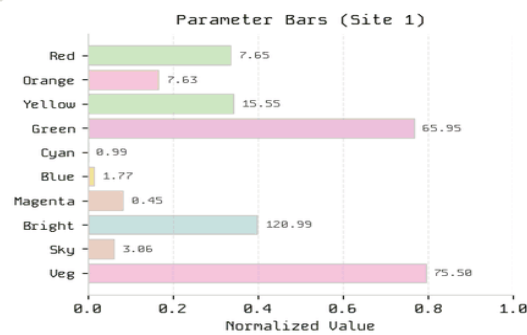
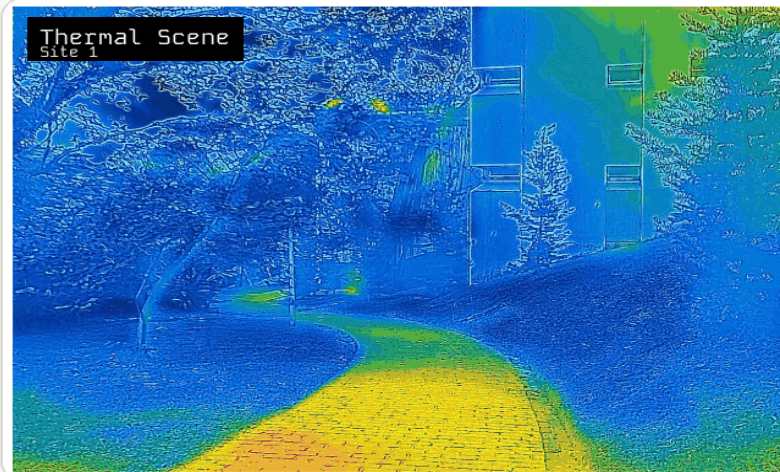


AI decarbonization model for Jurong Lake District

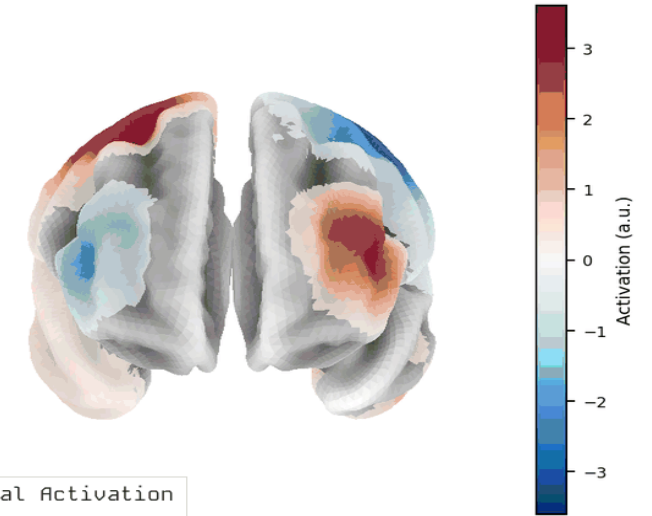


NUS Neuro-Perception AI for Urban Design | Site 1

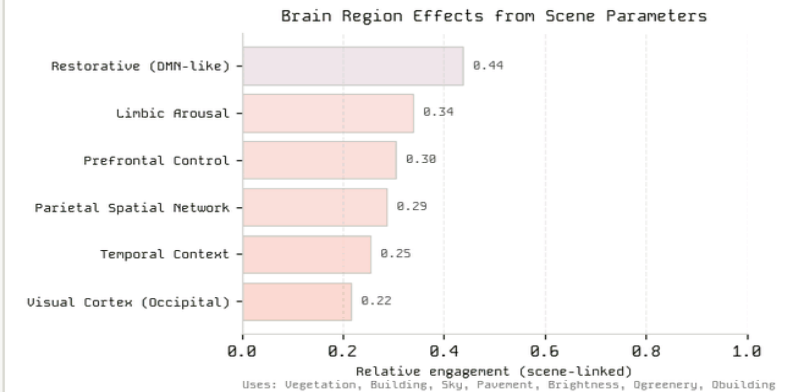
Lat: 1.296042 Long: 103.773331



Site 1 - Contrast Shift



Cortical Activation Site 1





URBANFLOW

AI Agents

Each a different expert

Energy Use Intensity
174 kWh/m²

Op Carbon
107 kgCO₂

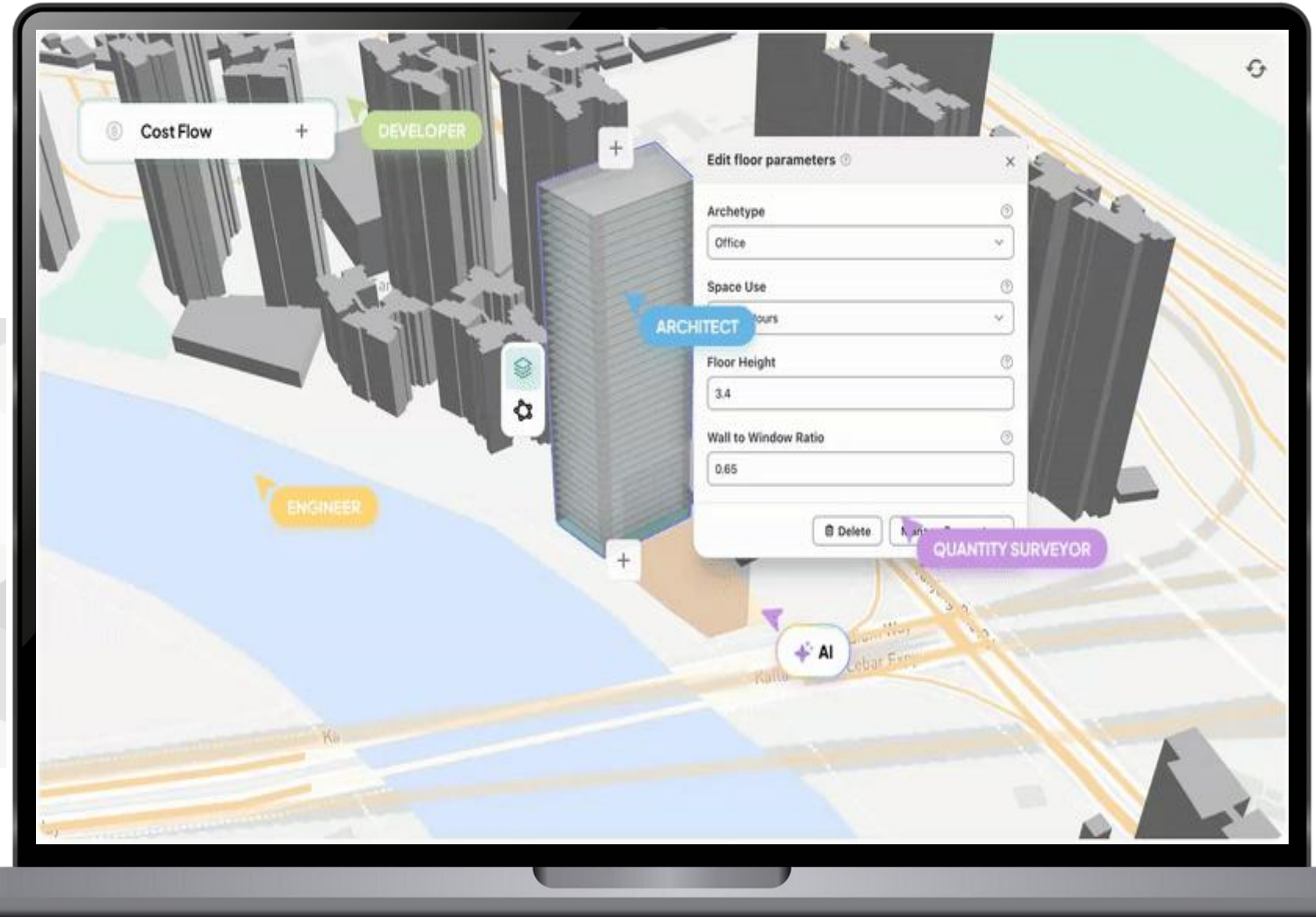
Insights

...consumes 33% of total energy during the day.
Introducing daylight sensors could reduce energy use

✦ Generating...

...propose and simulate energy saving measures for my current design|

✦ AI



Assessment

Class participation: 10%

Project: 90%

(Deliverables – Code or Repo + Slides + Conf. Paper)

No recordings (venue + logistical limitation)

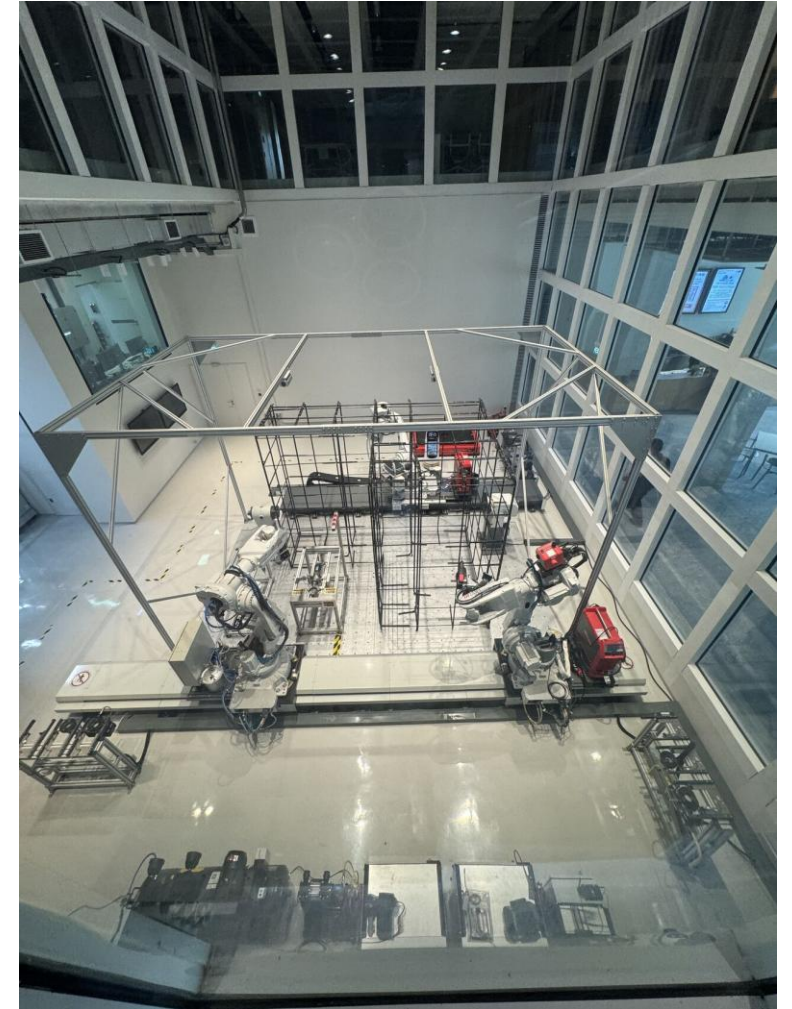
No attendance taking

No make-up classes

No weekly psets (hands-on lab + interim submissions)

Office hours will be announced

Site Visit



INTRODUCTION

BASIC

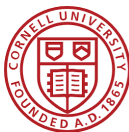
APPLICATIONS

BPS Alum



Pengdi LYu
Graduate Research
Assistant

Current Affiliation:
Cornell PhD



Cornell University



Kaifeng Zhu
Graduate Research
Assistant

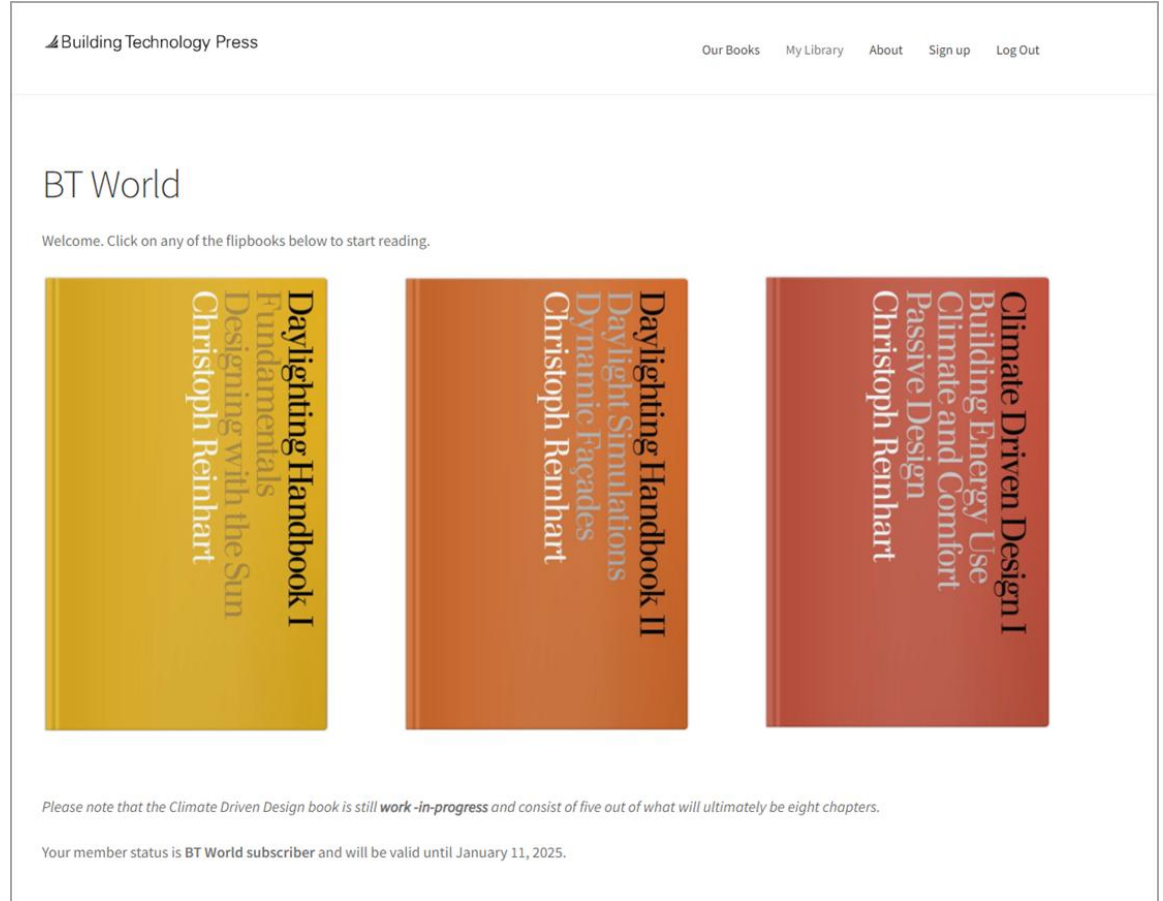
Current Affiliation: HKU
PhD



香港大學
THE UNIVERSITY OF HONG KONG



Software



Questions?

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L01.3

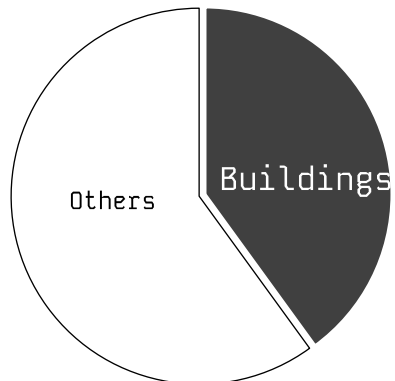
Applications

Examples

Design Beyond Space

~40% of energy use & of annual CO₂ emissions

Annual Global CO₂ Emissions



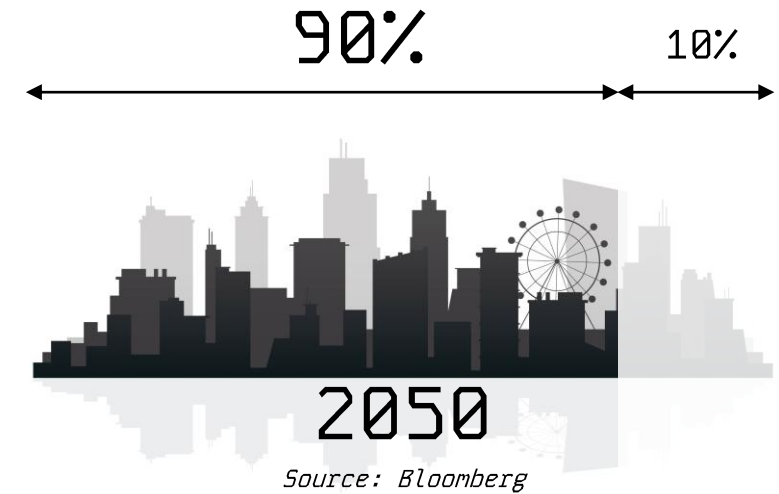
Source: UNEP

Global building floor area expected to **double** (230 billion m² new GFA)



Source: Architecture 2030

Around **90%** of existing buildings are expected to still be here in 2050







Source: Kelvin Ang Kah Eng



Source: Kelvin Ang Kah Eng





Universal drafting machine, combining T-square, triangles, scales, and protractor



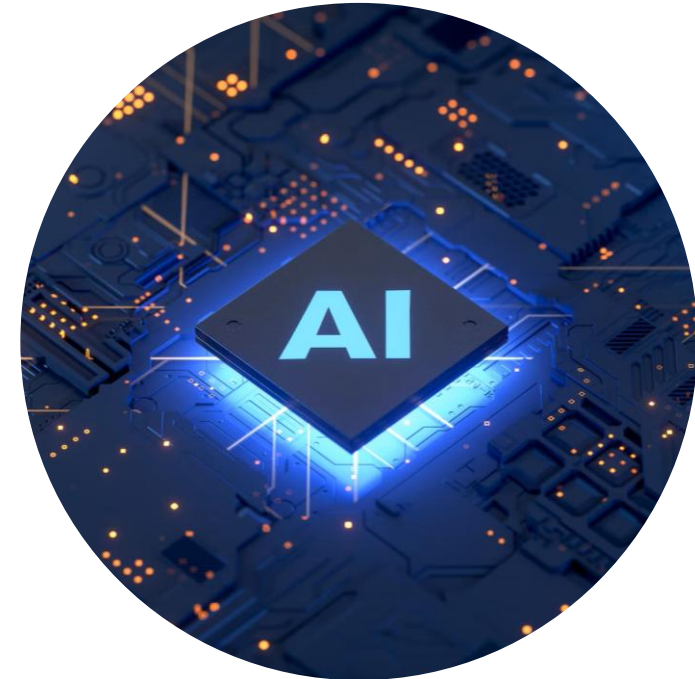
Old drafting table for architecture and engineering drafters to produce drawings



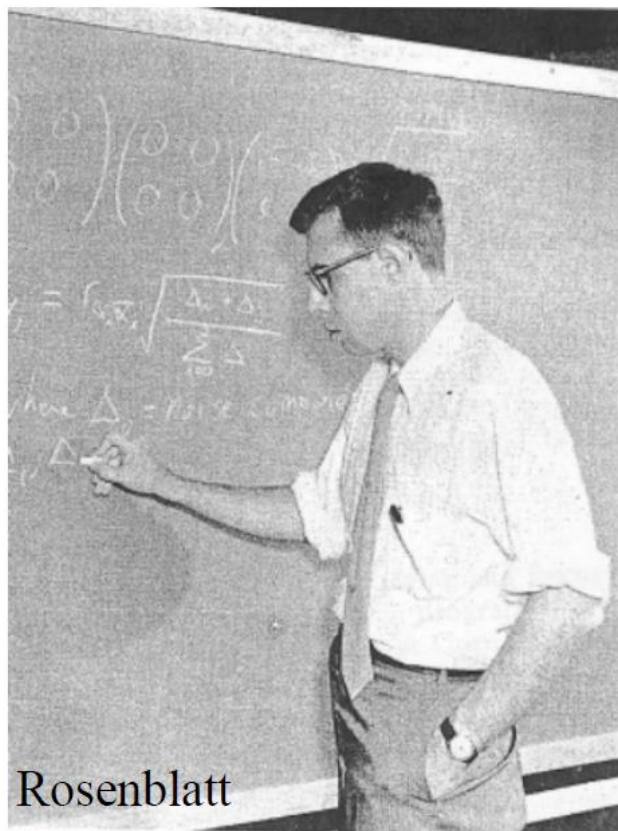
Calma Digitizer workstation introduced in 1965 allowed coordinate data to be entered and transferred to computers



One of the early generation of CAD systems



Perceptrons, 1958



Rosenblatt

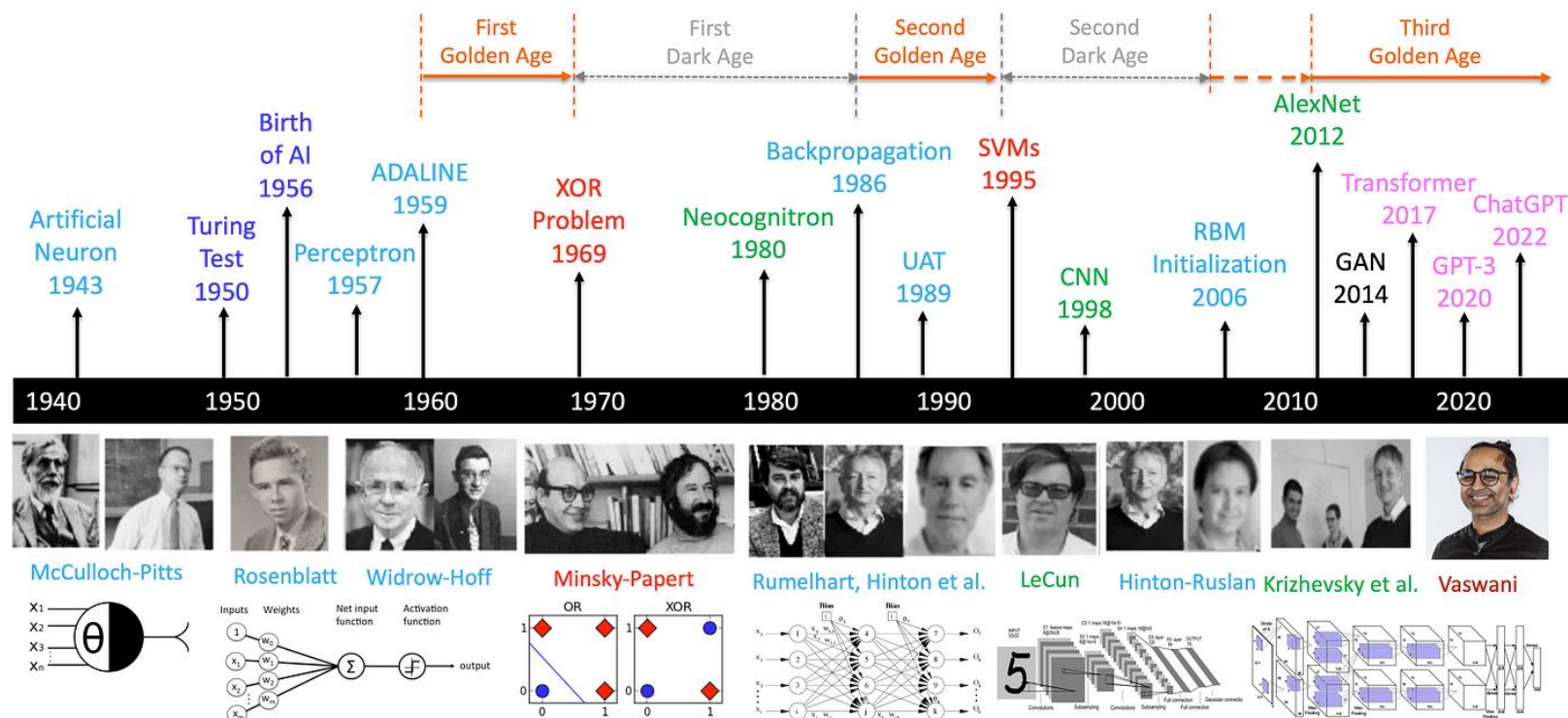
http://www.ecse.rpi.edu/homepages/nagy/PDF_chrono/2011_Nagy_Pace_FR.pdf. Photo by George Nagy

Neural Networks date back decades, so why now?

1. Big Data

2. Hardware

3. Software



ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



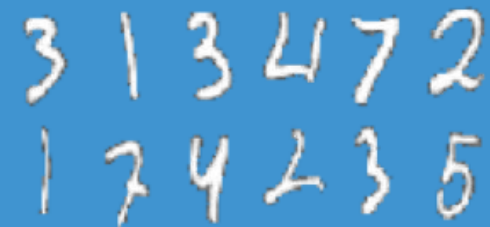
MACHINE LEARNING

Ability to learn without explicitly being programmed



DEEP LEARNING

Extract patterns from data using neural networks



Arthur Samuel (1959): Machine Learning is the field of study that gives computers the **ability to learn** without being explicitly programmed.

Goodfellow et al. (2016): Machine Learning is the ability of systems to acquire their own knowledge (as opposed to relying on hard-coded knowledge) by **extracting pattern from raw data**.

Tom Mitchell (1998): A computer program is said to learn from **experience E** with respect to some **task T** and some **performance measure P** , if its performance on T , as measured by P , improves with experience E .

Example: Your email program watches you label emails as spam or not spam, and based on that learns how to better filter spam.

E : You labeling them as spam or not

T : Classifying emails as spam or not

P : Fraction of emails correctly classified

Simple Definition: Machine Learning is the science of getting computers to learn without being explicitly programmed.

Training Data Setup: We have observed values of input variables X along with the output variable

Example: $X=(\text{Seniority}, \text{Years of Education})$, $Y=\text{Income}$

Denote the observations by $i=1,2,\dots,n$

Suppose the input data is p -dimensional

In our example: $p=2$, $n=\text{number of people in the dataset}$

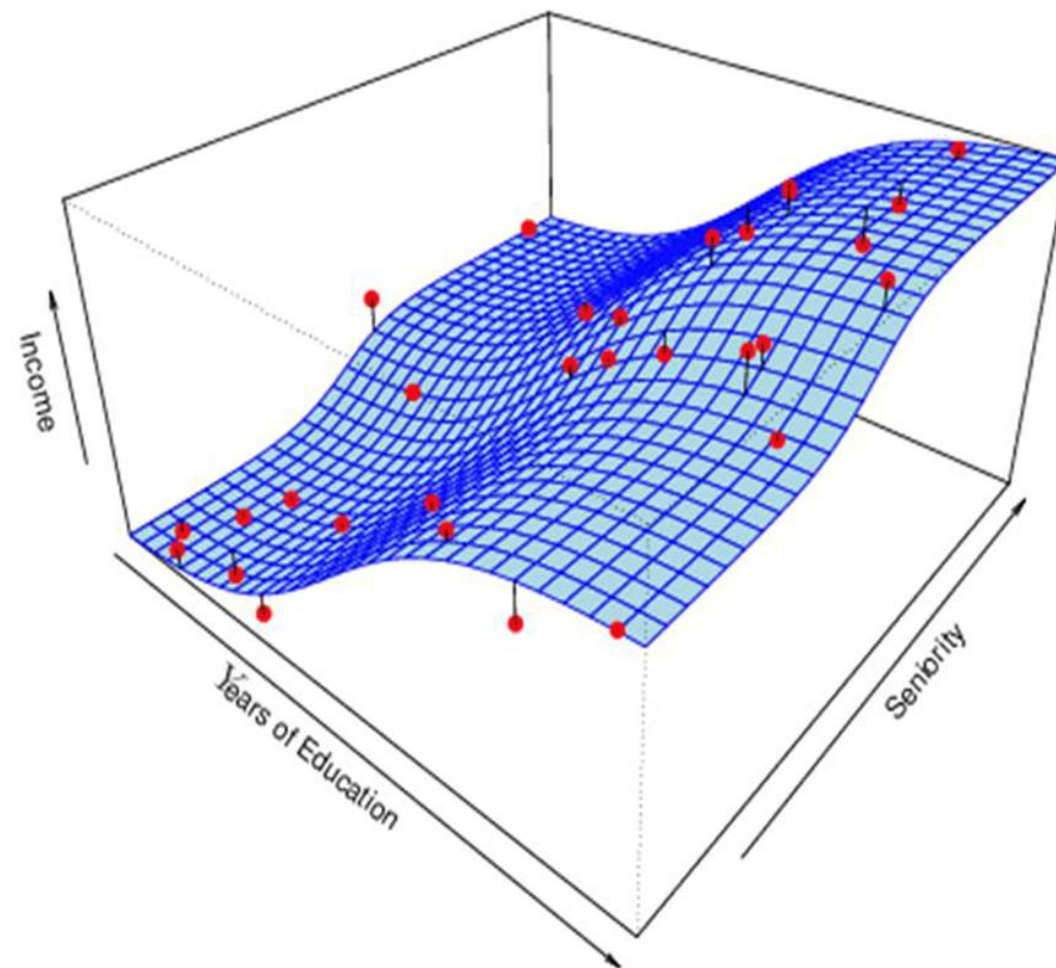
Observation $i(i=1,2,\dots,n): X_i=(X_{i1}, X_{i2}, \dots, X_{ip})$

We believe there is a relationship between Y and X .

$$\text{Income} = f(\text{Seniority}, \text{Education}) + \varepsilon$$

f represents the **blue surface**.

Red points represent the observed data points.



Why do we want to find f ?

Two main reasons:

Prediction: We can use f to predict values of Y for new value of X (income of a new person).

Inference: We might be interested in the type of relationship between Y and the X 's.

Which predictors affect the response (e.g., what affects income)?

- Is the relationship positive or negative?
- Is the relationship statistically significant?
- Is the relationship a simple linear one, or more complicated?

Our definitions of the two are different.

Machine Learning arose as a branch of Artificial Intelligence in Computer Science.

Statistical Learning arose as a branch of Statistics.

Machine Learning has a greater emphasis on **large-scale** and **real-time data** as well as **prediction accuracy** (sometimes treated as black boxes)

Statistical Learning has a greater emphasis on **model interpretability** and **statistical properties of estimation**.

However, the distinction is not significant now.

Supervised Learning in general

$S_n = \{(\mathbf{x}^{(t)}, \mathbf{y}^{(t)}), t = 1, \dots, n\}$ = dataset with features and labels (ground truth examples)

$\mathcal{H}$ = hypothesis class (space of functions)

$\mathcal{L}$ = loss function (often squared error for regression) e.g. $\mathcal{L}(x_1, x_2) = (x_1 - x_2)^2$

Goal: pick a function $h \in \mathcal{H}$ that predicts labels y based on features x as accurately as possible

>>> Optimization

ML algorithms always involve optimization (sometimes greedy). Formally:

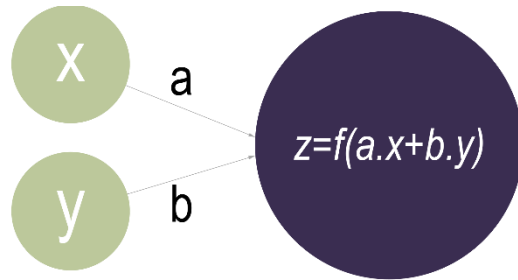
$$\min_{h \in \mathcal{H}} \sum_{i=1}^n \mathcal{L}(h(x_i), y_i)$$

prediction observation

Minimization over training data but we really want to do well over testing data!
(more on this next weeks)

“Giant Regression” = Universal Function Approximator

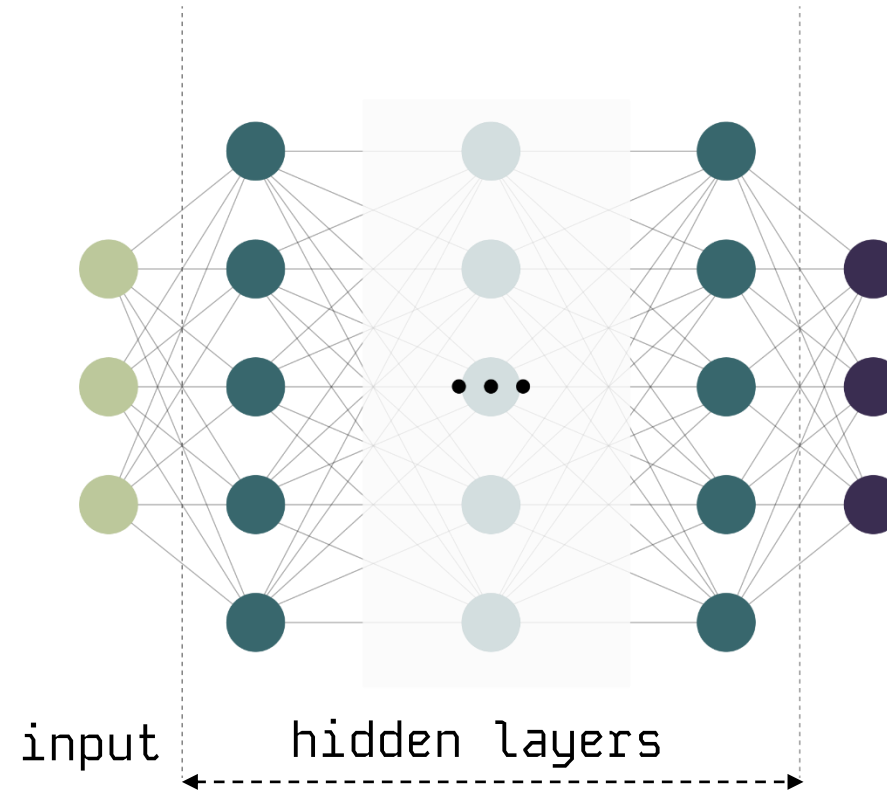
Neural Network



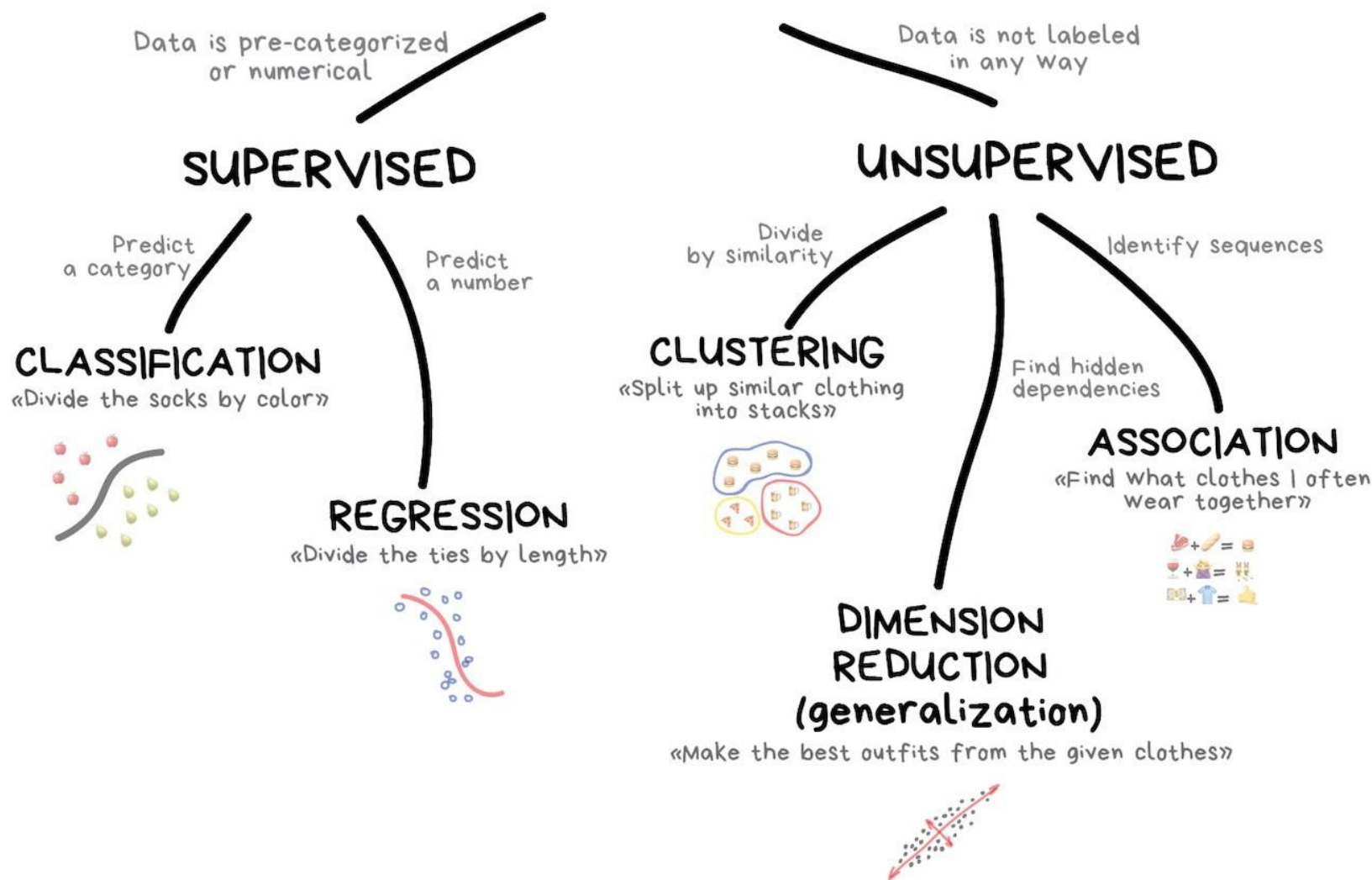
observation

$$\min \sum (z - z^*)$$

prediction



CLASSICAL MACHINE LEARNING



Supervised Learning

Both predictors (X) and response/outcome (Y) are observed or measured

Unsupervised Learning

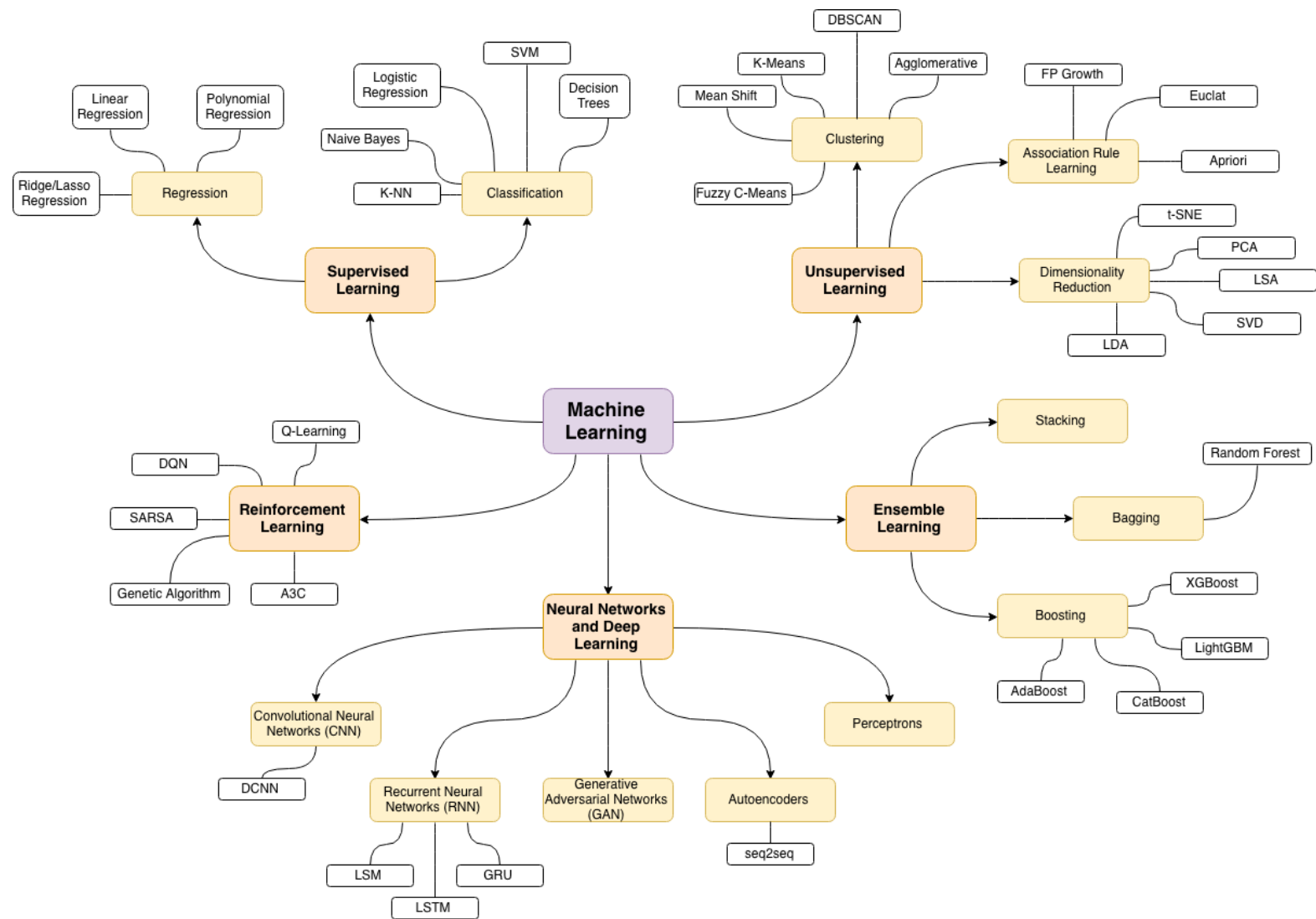
Only X values are available in the dataset; no output measure

Regression

Y is continuous/numerical
Example: Y = Income

Classification

Y is discrete/categorical
Example: Y = Spam or Not Spam



Transformers + Agents

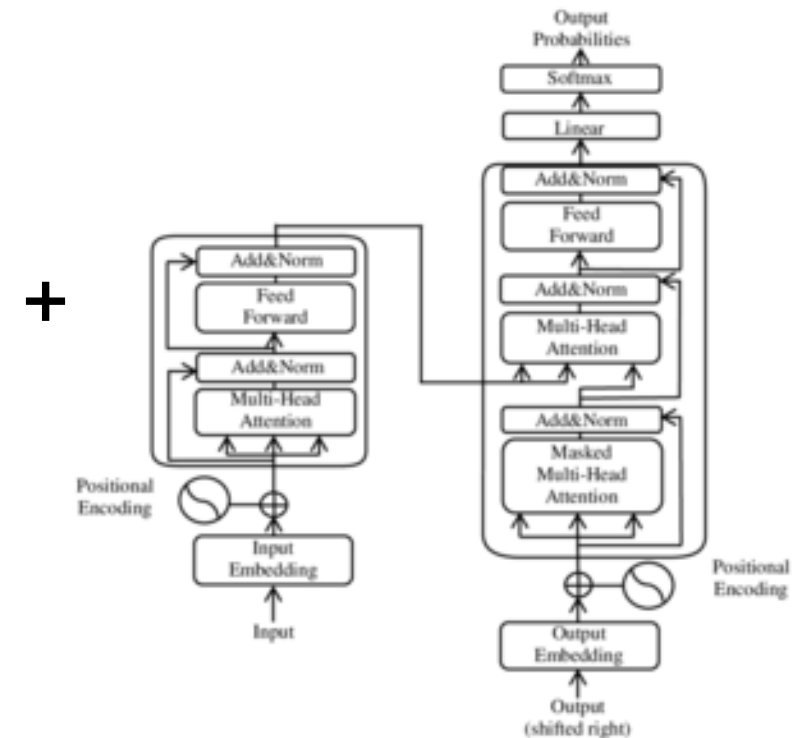


Image from Anil Jain, Google

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AI as a tool for designing buildings and cities

Effectively **analyze** complex data

Find **hidden patterns** in the data

Map set of **input variables** to **output variable**

Generate options

An Artwork Made by Artificial Intelligence Just Sold for \$400,000. I Am Shocked, Confused, Appalled.

By Jerry Saltz [@jerrysaltz](#)



Edmond de Belamy, from La Famille de Belamy. Photo: TIMOTHY A. CLARY/AFP/Getty Images

Last night Christie's did it again! The auction house sold an iffy object for an astronomical price.



How can AI help in Sustainable Building Design?

Where AI fits in the building lifecycle?

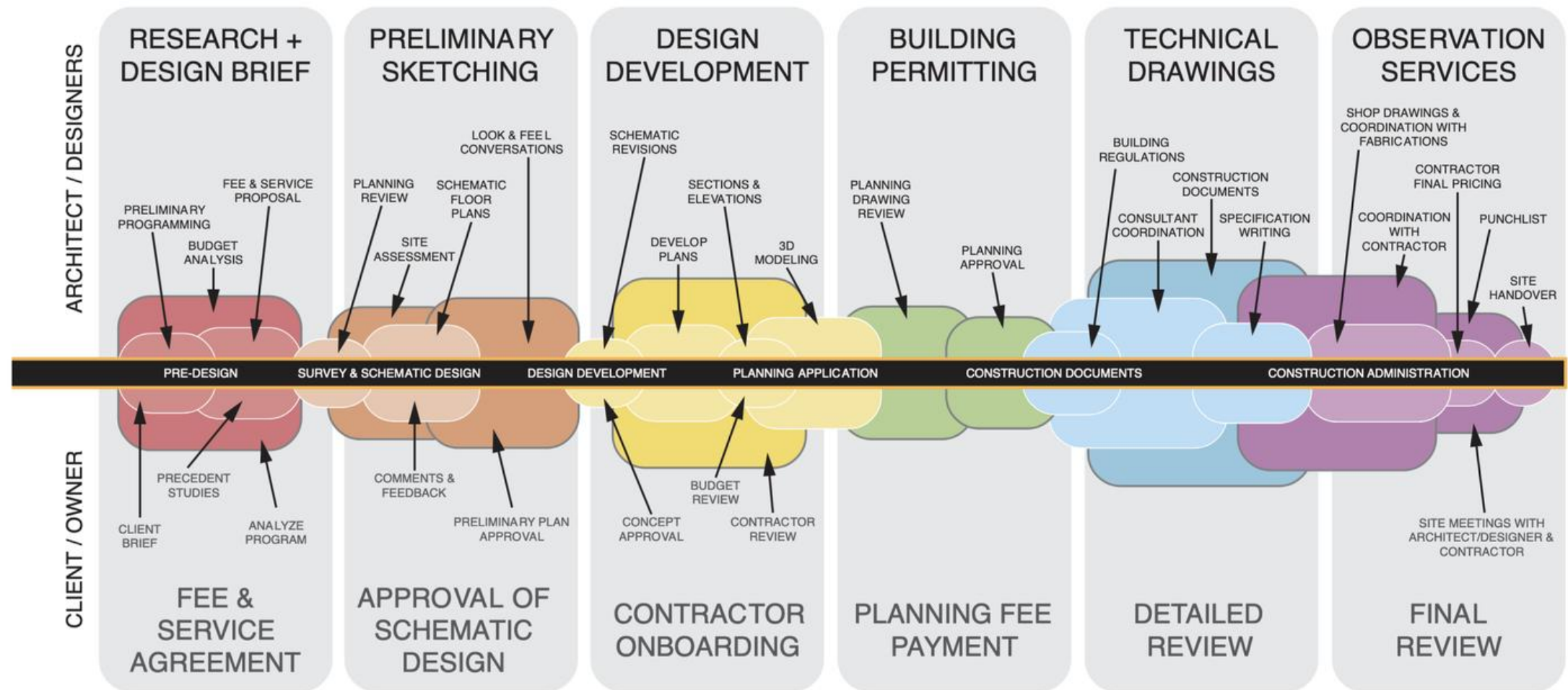


Image from Shahram Seifcar



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PROJECT

Using AI to Optimize Construction Design

Architecture, Engineering, and Construction

[Share page](#)

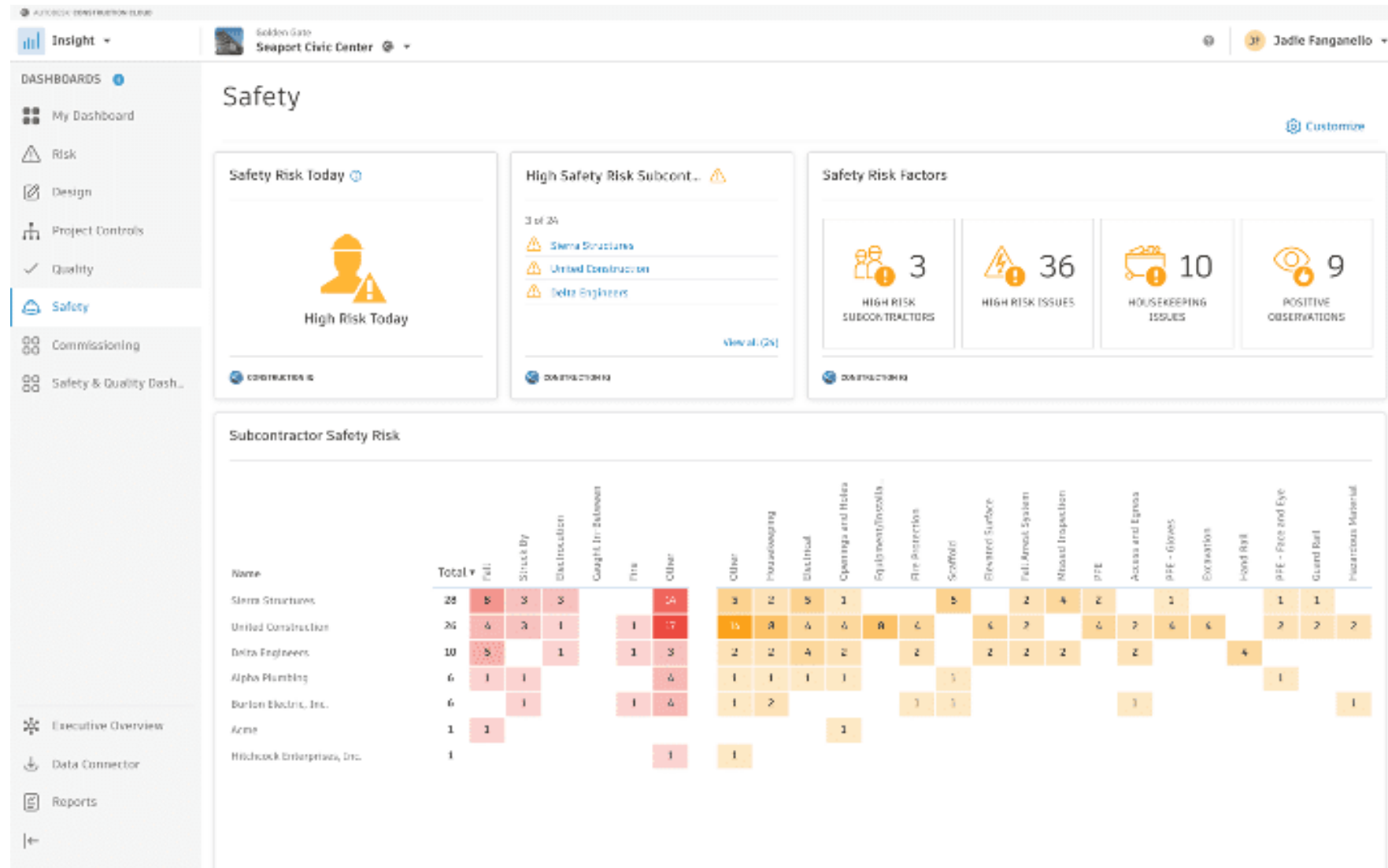


INTRODUCTION

BASICS

APPLICATIONS

Examples – Improve Safety



Examples – Regulatory Compliance

The screenshot shows a web browser window with the URL `dc-assistant.vercel.app/chat/DQyGzPP`. The interface includes a sidebar on the left for chat management and a main chat area on the right.

Chat History (Left Sidebar):

- Buttons: + New Chat, Wrong answer?
- History items:
 - What is the minimum road...
 - What guidelines govern the pro...
 - What is the minimum road setb...
 - What guidelines govern the pro...
- Clear history button at the bottom.

Chat Conversation (Main Area):

User: What is the minimum road setback for a 12-storey Condominium Development, along a Category 2 Road?

Assistant: For a 12-storey condominium development along a Category 2 road, the minimum road setback is 15 meters, which includes a 5-meter green buffer.

For more detailed information, you can refer to the [URA guidelines on Building Setback from Boundary for Flats and Condominiums](#).

References:

[Flats and Condominiums: Building Setback from Boundary](#) (15 Feb 2023)

Input Field: Send a message.

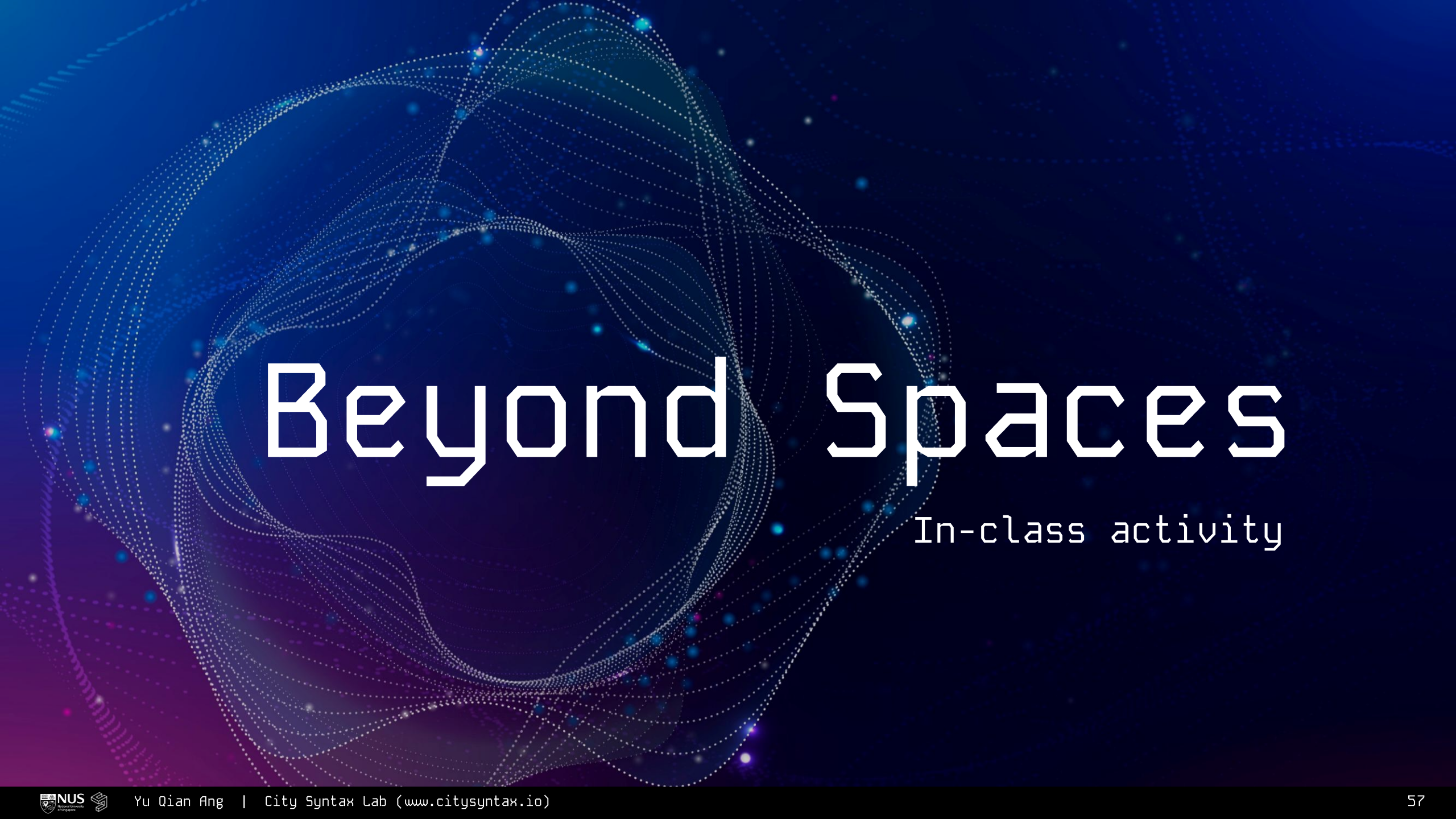
Footer Note: Use this chat assistant for workloads classified Official (Open) only. Double check responses for mistakes.

Go to:
<https://notebook.google.com/>

Go to:
<https://education.github.com/pack>

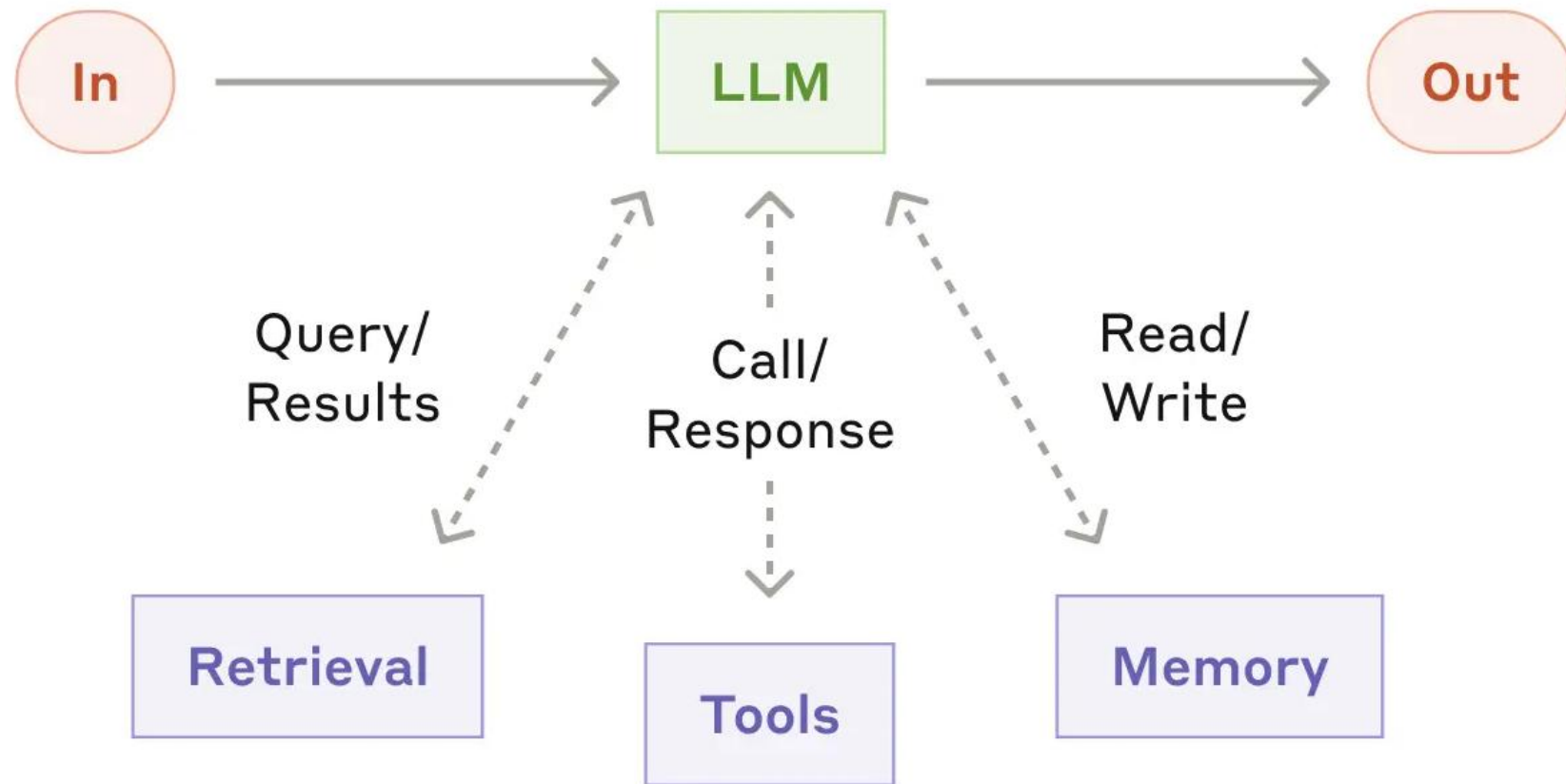
AI / LLM as we know it?

<https://bit.ly/BPS5231-transformers>

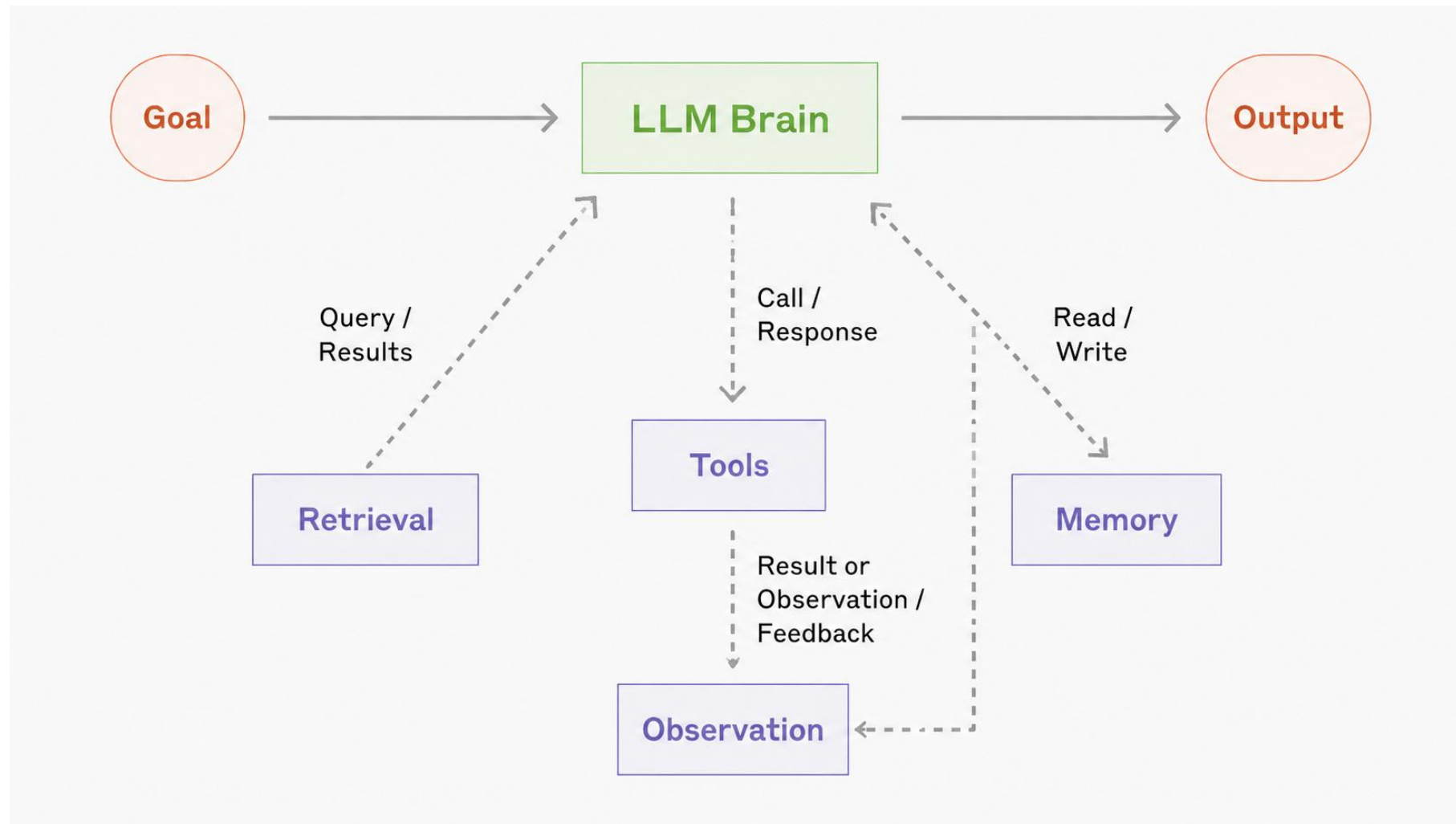


Beyond Spaces

In-class activity

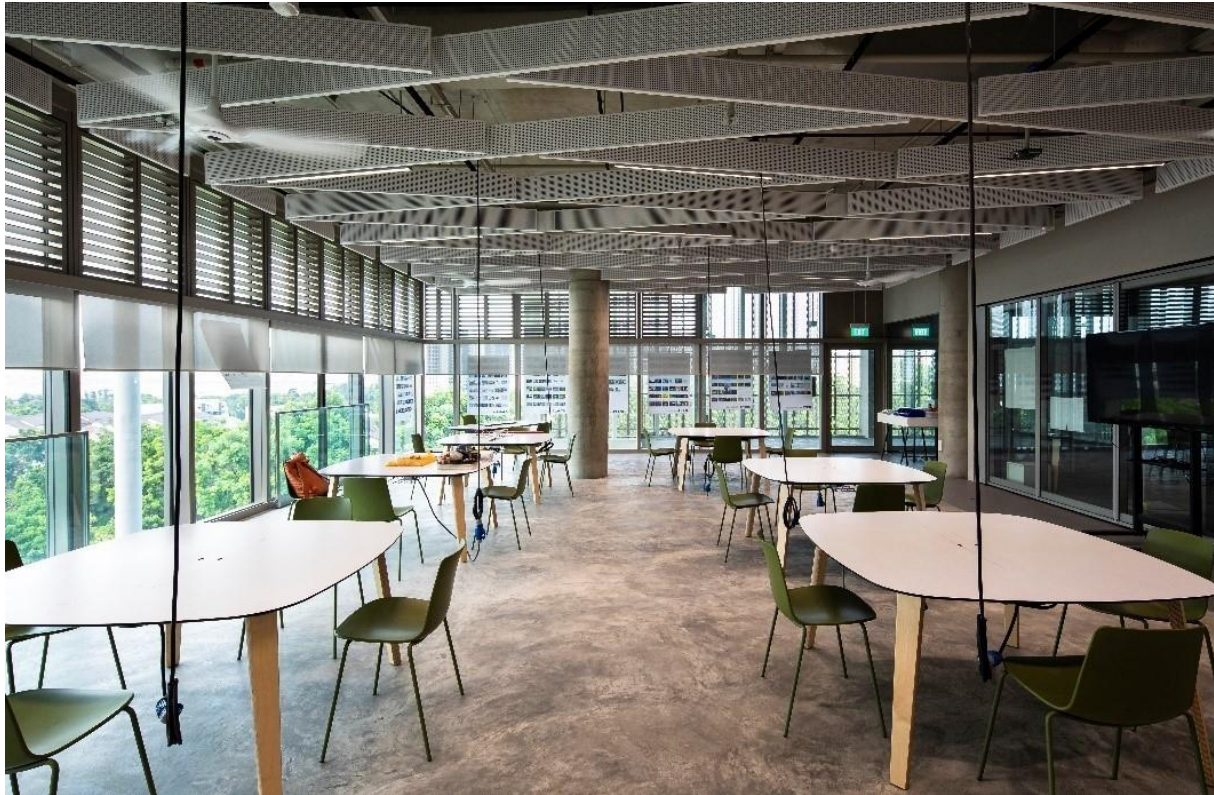


Source: Anthropic



Source: Anthropic

Choose a space in SDE, take a photo and change something you want



ChatGPT

Go to:
[https://bit.ly/BPS5231-L1-
BeyondSpaces](https://bit.ly/BPS5231-L1-BeyondSpaces)

